ROSPIN-SAT-1

Spacecraft Optical Analysis

Numerical Estimation of the Effective Solar Reflectance Factor

Numerical Optical Model and
Monte Carlo Directional Analysis

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1 Introduction

This report presents a method to calculate an effective solar reflectance factor, q, taking in consideration the optical properties and geometry of the spacecraft external surfaces. This coefficient can later be used in a model that calculates the radiation pressure.

$$F = \frac{F_s}{c} \cdot A_s \cdot (1 + q) \cdot \cos(I) \tag{1.1}$$

F_s is the incident solar irradiance, c is the speed of light, A_s is the reference illuminated area, I is the solar incidence angle, and q is the effective solar reflectance factor.

Having a solar propagation direction $\hat{\mathbf{k}}_\mathbf{i}$, the model returns an effective spacecraft reflectance $q_{\text{eff}}(\hat{\mathbf{k}}_\mathbf{i})$. Since the wave propagation direction is unknown, a fixed $\hat{k}_i$ vector direction is not assumed. Considering this, a Monte Carlo Simulation approach is used to analyse the direction dependence of q_{eff}. This method consists in generating pseudo random solar directions uniformly over a unit sphere.

2 Optical Reflectance Physics Model

Consider an electromagnetic wave that is incident on a plane, Σ. The incident medium is considered to be a vacuum with $n_1 \approx 1$. In order to represent the second medium, which coincides with the spacecraft surface material, the complex refractive index is used:

$$\tilde{n}_2(\lambda) = n_2(\lambda) + i\kappa(\lambda) \tag{2.1}$$

where n_2 is the real refractive index and κ is the extinction coefficient.

The incident wave vector is $\hat{\mathbf{k}}_i$, the reflected wave vector is $\hat{\mathbf{k}}_r$, and the transmitted wave vector is $\hat{\mathbf{k}}_t$. The outward unit normal to the surface is $\hat{\mathbf{n}}$.

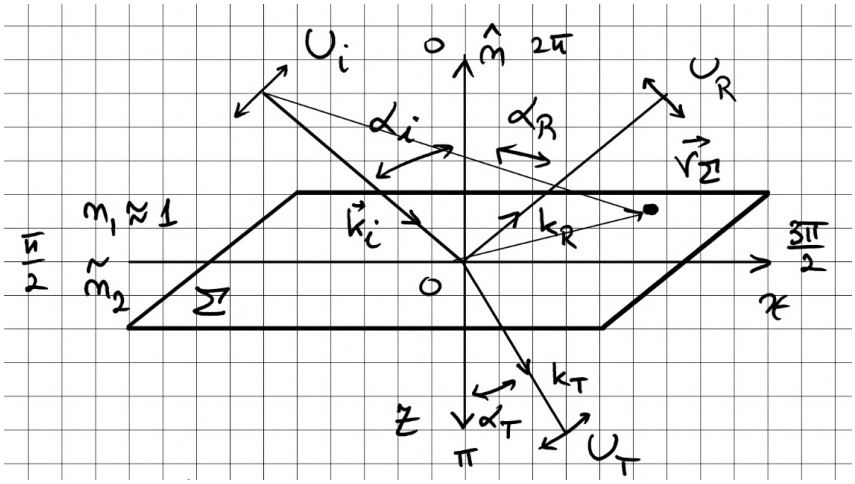


Figure 1: Geometry of reflection and transmission at a planar material interface, showing the incident, reflected and transmitted wave vectors and the corresponding angles.

The angle of incidence $\alpha_I = \angle(\hat{n}, -\hat{k}_i)$. The cosine of this angle can be used to determine if a face is illuminated or not. Therefore:

$$-\hat{\mathbf{k}}_i \cdot \hat{\mathbf{n}} = \|\hat{k}_i\| \cdot \|\hat{n}\| \cdot \cos(\alpha_I) = \cos(\alpha_I) \tag{2.2}$$

A surface is illuminated only when $\cos(\alpha_I) > 0$.

According to the law of reflection $\alpha_I = \alpha_R \equiv \alpha$, while $\alpha_T \equiv \beta$ is connected to α_I through Snell's law $n_1 \cdot sin(\alpha) = \tilde{n_2} \cdot sin(\beta)$.

$$cos(\beta) = \sqrt{1 - \left(\frac{sin(\alpha)}{\tilde{n}_2(\lambda)}\right)^2} \tag{2.3}$$

2.1 The reflectance general formula using the radiant flux

The radiant power incident on a planar surface can be obtained from the time-averaged Poynting vector.

$$\overline{\Phi} = \iint_S \langle \mathbf{S_P} \rangle \cdot \hat{\mathbf{n}}\, dS = \|\langle \mathbf{S_P} \rangle\| \cdot \|\hat{\mathbf{n}}\| \cdot cos(\alpha) \iint_S dS = I \cdot S \cdot cos(\alpha) \tag{2.4}$$

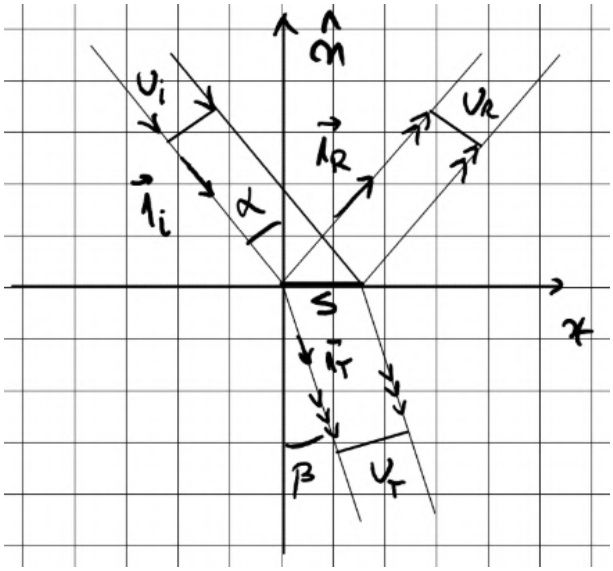


Figure 2: Geometrical interpretation of the radiant-flux surface integral at a planar interface.

The reflectance coefficient is defined as $q = \dfrac{\overline{\Phi_R}}{\overline{\Phi_I}}$. Using $\overline{\Phi} = IS\cos(\alpha)$ and $\alpha_R = \alpha_I$, it follows that $q = \dfrac{I_R}{I_I}$. For a monochromatic component of the electromagnetic field, $I \propto |E|^2$, hence $q = \dfrac{|E_R|^2}{|E_I|^2}$.

$$\begin{cases} U_I : \mathbf{E}_I = \mathbf{A}\, e^{i(\mathbf{k}_I \cdot \mathbf{r}_\Sigma - \omega t)} \\ U_R : \mathbf{E}_R = \mathbf{R}\, e^{i(\mathbf{k}_R \cdot \mathbf{r}_\Sigma - \omega t)} \\ U_T : \mathbf{E}_T = \mathbf{T}\, e^{i(\mathbf{k}_T \cdot \mathbf{r}_\Sigma - \omega t)} \end{cases}$$

Therefore,

$$q = \frac{|\mathbf{R}|^2}{|\mathbf{A}|^2} \tag{2.5}$$

The above relation is applied independently at each wavelength λ, and the resulting monochromatic reflectance is subsequently weighted by the solar spectral irradiance.

The electric-field amplitude can be decomposed into two orthogonal components relative to the plane of incidence, $\mathbf{A} = \mathbf{A}_\parallel + \mathbf{A}_\perp$. Since the two components are orthogonal, $|\mathbf{A}|^2 = |\mathbf{A}_\parallel|^2 + |\mathbf{A}_\perp|^2$, and similarly $|\mathbf{R}|^2 = |\mathbf{R}_\parallel|^2 + |\mathbf{R}_\perp|^2$. Therefore,

$$q = \frac{|\mathbf{R}_\parallel|^2 + |\mathbf{R}_\perp|^2}{|\mathbf{A}_\parallel|^2 + |\mathbf{A}_\perp|^2}.$$

Defining $q_\parallel = \dfrac{|\mathbf{R}_\parallel|^2}{|\mathbf{A}_\parallel|^2}$ and $q_\perp = \dfrac{|\mathbf{R}_\perp|^2}{|\mathbf{A}_\perp|^2}$, for unpolarized radiation $|\mathbf{A}_\parallel|^2 = |\mathbf{A}_\perp|^2$, hence

$$\boxed{q = \frac{q_\parallel + q_\perp}{2}}. \tag{2.6}$$

2.2 Fresnel Coefficients

The Fresnel amplitude reflection coefficients are

$$r_\perp = \frac{R_\perp}{A_\perp} = -\frac{\sin(\alpha-\beta)}{\sin(\alpha+\beta)}, \quad r_\parallel = \frac{R_\parallel}{A_\parallel} = \frac{\tan(\alpha-\beta)}{\tan(\alpha+\beta)},$$

The polarization-dependent reflectances are defined as

$$q_\parallel = \frac{I_R^{\parallel)}}{I_I^{(\parallel)}} = \frac{|R_\parallel|^2}{|A_\parallel|^2}, \qquad q_\perp = \frac{I_R^{(\perp)}}{I_I^{(\perp)}} = \frac{|R_\perp|^2}{|A_\perp|^2}.$$

Therefore,

$$q_\parallel = \left|\frac{\tan(\alpha-\beta)}{\tan(\alpha+\beta)}\right|^2, \quad q_\perp = |r_\perp|^2 = \left|\frac{\sin(\alpha-\beta)}{\sin(\alpha+\beta)}\right|^2.$$

Using Snell's law, the Fresnel coefficients can be rewritten in an equivalent form that avoids the indeterminate $\frac{0}{0}$ form which happens at normal incidence.

$$r_\perp = \frac{n_1\cos(\alpha) - \tilde{n}_2\cos(\beta)}{n_1\cos(\alpha) + \tilde{n}_2\cos(\beta)}, \quad r_\parallel = \frac{\tilde{n}_2\cos(\alpha) - n_1\cos(\beta)}{\tilde{n}_2\cos(\alpha) + n_1\cos(\beta)}.$$

Since the incident medium is a vacuum, $n_1 \simeq 1$, these reduce to

$$r_\perp = \frac{\cos(\alpha) - \tilde{n}_2\cos(\beta)}{\cos(\alpha) + \tilde{n}_2\cos(\beta)}, \quad r_\parallel = \frac{\tilde{n}_2\cos(\alpha) - \cos(\beta)}{\tilde{n}_2\cos(\alpha) + \cos(\beta)}.$$

Therefore,

$$\boxed{q(\alpha,\lambda) = \frac{1}{2}\left[\left|\frac{\tilde{n}_2(\lambda)\cos(\alpha) - \cos(\beta)}{\tilde{n}_2(\lambda)\cos(\alpha) + \cos(\beta)}\right|^2 + \left|\frac{\cos(\alpha) - \tilde{n}_2(\lambda)\cos(\beta)}{\cos(\alpha) + \tilde{n}_2(\lambda)\cos(\beta)}\right|^2\right]}. \tag{2.7}$$

where $\tilde{n}_2(\lambda) = n_2(\lambda) + i\kappa(\lambda)$, and, from Snell's law, $\cos(\beta) = \sqrt{1 - \left(\dfrac{\sin(\alpha)}{\tilde{n}_2(\lambda)}\right)^2}$.

Equation (2.7) gives the monochromatic reflectance of the considered surface as a function of the incidence angle and wavelength.

2.3 Solar Spectral Weighting

The solar radiation is not monochromatic. Its spectral distribution is described by the solar spectral irradiance, $I_\odot = I_\odot(\lambda) \quad \left[\frac{\mathrm{W}}{\mathrm{m}^2\,\mathrm{nm}}\right]$

The total solar irradiance over the considered wavelength interval is

$$I_{\odot,\mathrm{tot}} = \int_{\lambda_{\min}}^{\lambda_{\max}} I_\odot(\lambda)\, d\lambda.$$

For a surface of area S illuminated at an incidence angle α, the incident radiant power is

$$\Phi_\odot^I = S\cos(\alpha) \int_{\lambda_{\min}}^{\lambda_{\max}} I_\odot(\lambda)\, d\lambda.$$

At each wavelength, the reflected spectral contribution is

$$q_\odot(\alpha,\lambda) = \frac{\overline{\Phi}_\odot^R}{\overline{\Phi}_\odot^I} \Rightarrow \overline{\Phi}_\odot^R = q_\odot(\alpha,\lambda)\cdot\overline{\Phi}_\odot^I \Rightarrow d\overline{\Phi}_\odot^R = q_\odot(\alpha,\lambda)\cdot d\overline{\Phi}_\odot^I$$

$$d\overline{\Phi}_\odot^R = q(\alpha,\lambda)\; I_\odot(\lambda)\; S\cos(\alpha)\; d\lambda.$$

Therefore, the total reflected radiant power is

$$\overline{\Phi}_\odot^R = S\cos(\alpha) \int_{\lambda_{\min}}^{\lambda_{\max}} q(\alpha,\lambda)\, I_\odot(\lambda)\, d\lambda.$$

which gives

$$\boxed{q_\odot(\alpha) = \frac{\displaystyle\int_{\lambda_{\min}}^{\lambda_{\max}} q(\alpha,\lambda)\, I_\odot(\lambda)\, d\lambda}{\displaystyle\int_{\lambda_{\min}}^{\lambda_{\max}} I_\odot(\lambda)\, d\lambda}} \tag{2.8}$$

Here, $q(\alpha,\lambda)$ is the monochromatic reflectance obtained from the Fresnel model, while $I_\odot(\lambda)$ acts as the spectral weighting function.

Using equations (2.1), (2.3), (2.7) and (2.8) a complete optical model was built for a single surface.

3 Spacecraft Effective Reflectance

After writing a mathematical model that correctly returns the effective solar reflectance for one surface, the most natural thing is to extend this model to M surfaces. For each spacecraft surface j, the optical-geometrical model is defined as

$$\mathcal{S}_j = \left\{S_j,\, \hat{\mathbf{n}}_j,\, \tilde{n}_j(\lambda)\right\}_{j=1}^{N},$$

where S_j is the surface area, $\hat{\mathbf{n}}_j$ is the outward unit normal, and $\tilde{n}_j(\lambda)$ is the complex refractive index of the corresponding surface material. Because $n_1 \approx 1$, we can make the convention $n_2^j = n_j$. Similarly for $\tilde{n}_2^j$.

For each surface j, the incidence angle is obtained from $\cos\alpha_j = -\hat{\mathbf{k}} \cdot \hat{\mathbf{n}}_j$. Only surfaces satisfying $\cos\alpha_j > 0$ are illuminated and contribute to the reflected radiant power. The material of surface j is described by the complex refractive index

$$\tilde{n}_j(\lambda) = n_j(\lambda) + i\kappa_j(\lambda). \tag{3.1}$$

The monochromatic reflectance of surface j is therefore

$$q_j(\alpha_j, \lambda) = \frac{1}{2}\left[\left|\frac{\tilde{n}_j(\lambda)\cos(\alpha_j) - \cos(\beta_j)}{\tilde{n}_j(\lambda)\cos(\alpha_j) + \cos(\beta_j)}\right|^2 + \left|\frac{\cos(\alpha_j) - \tilde{n}_j(\lambda)\cos(\beta_j)}{\cos(\alpha_j) + \tilde{n}_j(\lambda)\cos(\beta_j)}\right|^2\right], \tag{3.2}$$

with

$$\cos\beta_j = \sqrt{1 - \left(\frac{\sin(\alpha_j)}{\tilde{n}_j(\lambda)}\right)^2}. \tag{3.3}$$

The solar-weighted reflectance of surface j is

$$q_\odot^j(\alpha_j) = \frac{\displaystyle\int_{\lambda_{\min}}^{\lambda_{\max}} q_j(\alpha_j, \lambda) I_\odot(\lambda)\, d\lambda}{\displaystyle\int_{\lambda_{\min}}^{\lambda_{\max}} I_\odot(\lambda)\, d\lambda}. \tag{3.4}$$

For each illuminated surface j, the incident radiant power is

$$\overline{\Phi}_{I,j} = I_{\odot,\text{tot}}\, S_j \cos(\alpha_j),$$

while the reflected radiant power is

$$\overline{\Phi}_{R,j} = q_\odot^j(\alpha_j)\, \overline{\Phi}_{I,j} = q_\odot^j(\alpha_j)\, I_{\odot,\text{tot}}\, S_j \cos(\alpha_j).$$

The total incident and reflected radiant powers for the spacecraft are therefore

$$\overline{\Phi}_I = \sum_{j=1}^{N} \overline{\Phi}_{I,j} = \sum_{j=1}^{N} I_{\odot,\text{tot}}\, S_j \cos(\alpha_j),$$

$$\overline{\Phi}_R = \sum_{j=1}^{N} \overline{\Phi}_{R,j} = \sum_{j=1}^{N} q_\odot^j(\alpha_j) I_{\odot,\text{tot}}\, S_j \cos(\alpha_j).$$

The effective spacecraft reflectance is defined as

$$q_{\text{eff}} = \frac{\overline{\Phi}_R}{\overline{\Phi}_I}.$$

Substituting the expressions above and simplifying $I_{\odot,\text{tot}}$ we get

$$\boxed{q_{\text{eff}}(\hat{\mathbf{k}}_\mathbf{I}) = \frac{\sum_{j=1}^{N} q_\odot^j(\alpha_j) S_j \cos(\alpha_j)}{\sum_{j=1}^{N} S_j \cos(\alpha_j)}} \tag{3.5}$$

where only illuminated surfaces, satisfying $\cos\alpha_j > 0$, contribute to the summation.

The effective spacecraft reflectance is therefore a projected-area weighted average of the solar-weighted reflectances of the illuminated external surfaces.

4 Monte Carlo Simulation

Since no fixed solar illumination direction is assumed, the solar propagation vector is sampled uniformly over the unit sphere.

A uniform distribution over the sphere requires the probability density to be proportional to the spherical surface element $dS = \sin\theta\, d\theta\, d\phi$.

Introducing

$$u = \cos(\theta),$$

gives

$$du = -\sin(\theta)\, d\theta.$$

Therefore, a uniform distribution over the sphere can be obtained by sampling

$$u \sim \mathcal{U}[-1, 1], \qquad \phi \sim \mathcal{U}[0, 2\pi).$$

The corresponding solar propagation direction is

$$\boxed{\hat{\mathbf{k}}_\mathbf{I} = \begin{pmatrix} \sin(\theta)\cos(\phi) \\ \sin(\theta)\sin(\phi) \\ \cos(\theta) \end{pmatrix} = \begin{pmatrix} \sqrt{1-u^2}\cos(\phi) \\ \sqrt{1-u^2}\sin(\phi) \\ u \end{pmatrix}}$$

For each Monte Carlo realization t, a solar propagation direction $\hat{\mathbf{k}}_t$ is generated and the corresponding effective spacecraft reflectance is evaluated as $q_t = q_{\text{eff}}(\hat{\mathbf{k}}_t)$. The procedure is repeated for M samples.

The mean effective reflectance is

$$\bar{q} = \frac{1}{M} \sum_{t=1}^{M} q_t.$$

5 Input Data and Numerical Implementation

5.1 Optical Material Properties

The optical properties of the external spacecraft surfaces are defined using the complex refractive indices

$$\tilde{n}_j(\lambda) = n_j(\lambda) + i\kappa_j(\lambda).$$

For Al surfaces, the optical data are taken from the Rakić dataset. The current simulation treats all Al surfaces using the same bulk Al optical model. For the CMG-100 coverglass, a simplified optical model is adopted using $n_2 = 1.516$ and $\kappa = 0$.

5.2 Solar Spectrum

The incident solar radiation is represented using the ASTM E490 AM0 solar spectral irradiance. The spectral integration is performed over the interval 250 nm $\leq \lambda \leq$ 2500 nm. The spectrum is numerically integrated using the trapezoidal rule.

5.3 Spacecraft Surface Model

Table 1: External spacecraft surfaces included in the optical model.

Surface	**Material**	**Area [mm^2]**	**Normal**
Main Back Plate	Al-6061-T6	4731.2235	$+\hat{\mathbf{y}}$
Back Plate (COM)	Al-5005-H14	7380.0363	$+\hat{\mathbf{y}}$
Shield (COM)	Al-6082-T651	6361.2203	$+\hat{\mathbf{y}}$
Side Rail	Al-7075-T6	5077.5000	$+\hat{\mathbf{y}}$
OAB	Al-6061-T6	1977.1523	$+\hat{\mathbf{y}}$
Side Rail	Al-7075-T6	5077.5000	$+\hat{\mathbf{x}}$
OAB	Al-6061-T6	1977.1523	$+\hat{\mathbf{x}}$
Fixed Solar Cells (FSC)	CMG-100 Glass Cover	21124.2500	$+\hat{\mathbf{x}}$
Deployable Solar Cells (DSC)	CMG-100 Glass Cover	36213.0000	$+\hat{\mathbf{x}}$
Back Plate (FSC)	Al-5005-H14	3935.8699	$+\hat{\mathbf{x}}$
Back Plate (DSC)	Al-6082-T651	10208.4726	$+\hat{\mathbf{x}}$
Side Rail	Al-7075-T6	5077.5000	$-\hat{\mathbf{y}}$
OAB	Al-6061-T6	1977.1523	$-\hat{\mathbf{y}}$
Back Plate	Al-6082-T651	23004.8134	$-\hat{\mathbf{y}}$
Side Rail	Al-7075-T6	5077.5000	$-\hat{\mathbf{x}}$
OAB	Al-6061-T6	1977.1523	$-\hat{\mathbf{x}}$
FSC	CMG-100 Glass Cover	21124.2500	$-\hat{\mathbf{x}}$
Back Plate (FSC)	Al-5005-H14	3935.8699	$-\hat{\mathbf{x}}$
Back Plate (DSC)	Al-6082-T651	46009.6268	$-\hat{\mathbf{x}}$
Chorus Shield	Al-6082-T651	8478.2977	$+\hat{\mathbf{z}}$
Top Frame + Camera Housing	Al-6061-T6	8629.5793	$-\hat{\mathbf{z}}$

For additional details, refer to Appendix A, *Spacecraft Surface Identification*.

5.4 Numerical Procedure

5.4.1 Solver Architecture

The numerical implementation is organized around a solver that receives the spacecraft surface definition, the material optical data, and the solar spectrum. Each spacecraft surface is represented by its area, outward unit normal and material identifier. Each material is associated with wavelength-dependent optical constants $n(\lambda)$ and $\kappa(\lambda)$.

The solver evaluates the optical model in three main stages:

1. evaluation of the monochromatic Fresnel reflectance;
2. solar spectral weighting for each illuminated surface;
3. projected-area weighting of all illuminated spacecraft surfaces.

For a given wavelength and incidence angle, the refractive index of the corresponding material is obtained by interpolation of the optical data. The Fresnel model is then used to evaluate $q_j(\alpha_j, \lambda)$. The solar-weighted reflectance $q_{\odot}^{j}(\alpha_j)$ is evaluated numerically over the solar spectrum using trapezoidal integration. Finally, for a given solar propagation direction $\hat{\mathbf{k}}_{\mathbf{I}}$, the solver evaluates all illuminated surfaces and computes the effective spacecraft reflectance using the projected-area weighted relation derived previously.

To reduce the computational cost of the Monte Carlo simulation, the material refractive indices are pre-evaluated on the solar spectral grid before the directional sampling is performed.

5.4.2 Monte Carlo Execution

For each Monte Carlo realization, two independent pseudo-random variables are generated,

$$u \sim \mathcal{U}[-1, 1], \qquad \phi \sim \mathcal{U}[0, 2\pi),$$

from which the solar propagation direction is constructed.

The solver then evaluates $q_{\text{eff}}(\hat{\mathbf{k}})$ and the corresponding projected illuminated area. The procedure is repeated for $M = 100000$ samples. A fixed random seed is used to ensure reproducibility.

5.5 Source Code Availability

The numerical implementation used for the optical and Monte Carlo analysis is available in the project repository: https://github.com/sorin373/Effective-Solar-Reflectance-Factor.

6 Numerical Verification

The numerical implementation was verified through a set of consistency and convergence checks. The verification procedure focuses on the correct evaluation of the optical data, spectral integration, spacecraft geometry and Monte Carlo sampling.

6.1 Input Data and Interpolation Checks

The optical solver performs several internal consistency checks during data loading and interpolation.

The wavelength-dependent optical properties are loaded from tabulated data and linearly interpolated to the wavelengths of the solar spectral grid. The implementation verifies that the requested wavelength lies inside the available optical-data range and reports an error if interpolation cannot be performed.

Similarly, the solar spectral data are restricted to the wavelength interval used by the analysis $250\ \text{nm} \leq \lambda \leq 2500\ \text{nm}$.

The solver also verifies that at least two spectral points are available and that the wavelength values are strictly increasing before numerical integration is performed.

6.2 Spectral Integration Consistency

The solar-weighted reflectance is evaluated numerically using the trapezoidal rule. For every wavelength interval $[\lambda_i, \lambda_{i+1}]$, both the reflected spectral contribution and the incident solar irradiance are integrated using the same quadrature method.

A zero denominator is explicitly detected by the solver and treated as an invalid solar-spectrum input.

6.3 Principal Illumination Directions

The geometrical implementation was checked by evaluating the spacecraft reflectance for the six principal illumination directions,

$$\pm X, \qquad \pm Y, \qquad \pm Z.$$

For each test case, the solar propagation vector is imposed directly and the effective spacecraft reflectance is evaluated.

6.4 Monte Carlo Convergence

The convergence of the Monte Carlo solution was evaluated by repeating the simulation with increasing numbers of uniformly distributed solar illumination directions. The investigated sample counts were $10^1, 10^2, 10^3, 10^4, 10^5$ with the same pseudo-random seed used in each case. For every sample count, the mean reflectance, standard deviation, 5th percentile, median, 95th percentile and standard error of the mean were evaluated.

Table 2: Monte Carlo convergence with increasing sample count.

M	Mean	Std. Dev.	P05	P50	P95	Std. Error
10	0.663325	0.146111	0.437228	0.719169	0.776447	0.046204
100	0.672647	0.166827	0.369331	0.725954	0.891654	0.016683
1000	0.668384	0.178664	0.344951	0.725954	0.907284	0.005650
10000	0.670150	0.177165	0.347265	0.724977	0.906998	0.001772
100000	0.667812	0.175791	0.348638	0.723773	0.904695	0.000556

The statistical quantities become progressively more stable as the number of Monte Carlo samples increases. The standard error of the mean decreases approximately according to

$$\mathrm{SE} = \frac{\sigma}{\sqrt{M}},$$

as expected for Monte Carlo sampling. For $M = 10^5$, $SE \approx 5.56 \times 10^{-4}$. The $M = 10^5$ simulation was therefore considered sufficiently converged for the final statistical analysis.

7 Results

7.1 Statistical Reflectance Results

The final Monte Carlo analysis was performed using $M = 10^5$ uniformly distributed solar illumination directions. The resulting effective spacecraft reflectance distribution is

$$\boxed{\bar{q} = 0.667812, \qquad \sigma_q = 0.175791.}$$

The selected distribution percentiles are

$$\boxed{P_{05} = 0.348638, \qquad P_{50} = 0.723773, \qquad P_{95} = 0.904695.}$$

Here, P_{05} and P_{95} represent the reflectance values below which approximately 5% and 95% of the sampled illumination directions fall, respectively, while P_{50} represents the median of the distribution.

Based on the mean of the directional reflectance distribution, the nominal effective solar reflectance factor is taken as

$$\boxed{q_{\text{nominal}} \approx 0.67} \tag{7.1}$$

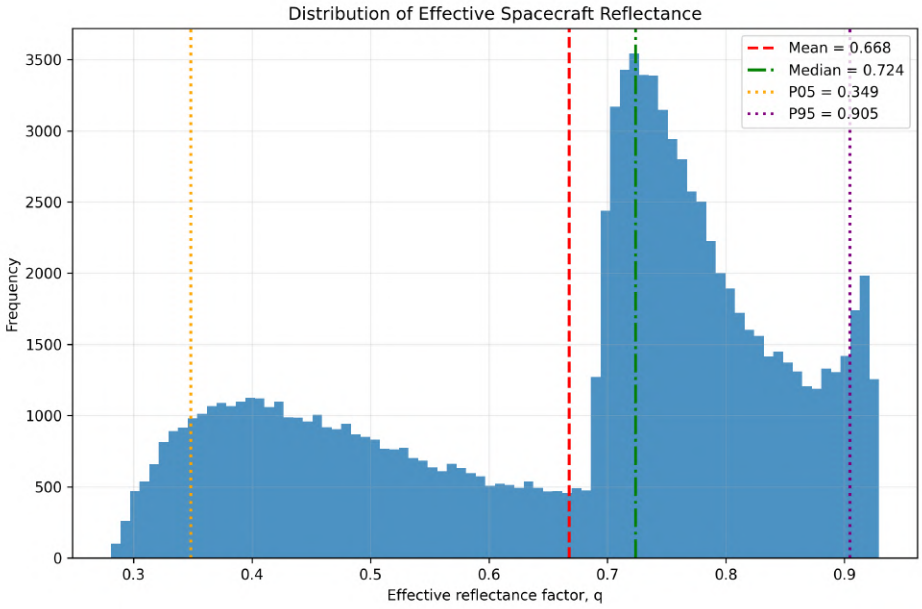


Figure 3: Histogram of the effective spacecraft reflectance factor obtained from the Monte Carlo simulation. Each bar represents the number of sampled illumination directions producing a reflectance value within the corresponding interval.

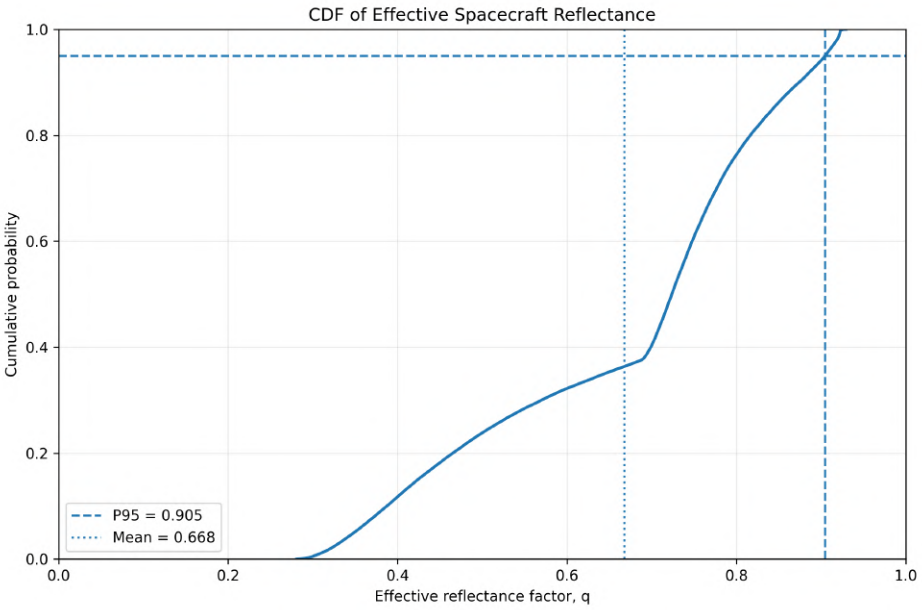


Figure 4: Empirical cumulative distribution of the effective spacecraft reflectance obtained from the Monte Carlo simulation.

7.2 Principal Illumination Directions

Table 3: Effective reflectance for principal illumination directions.

Illuminated Face	q_{eff}
$+X$	0.280207
$-X$	0.685767
$+Y$	0.924324
$-Y$	0.924324
$+Z$	0.924324
$-Z$	0.924324

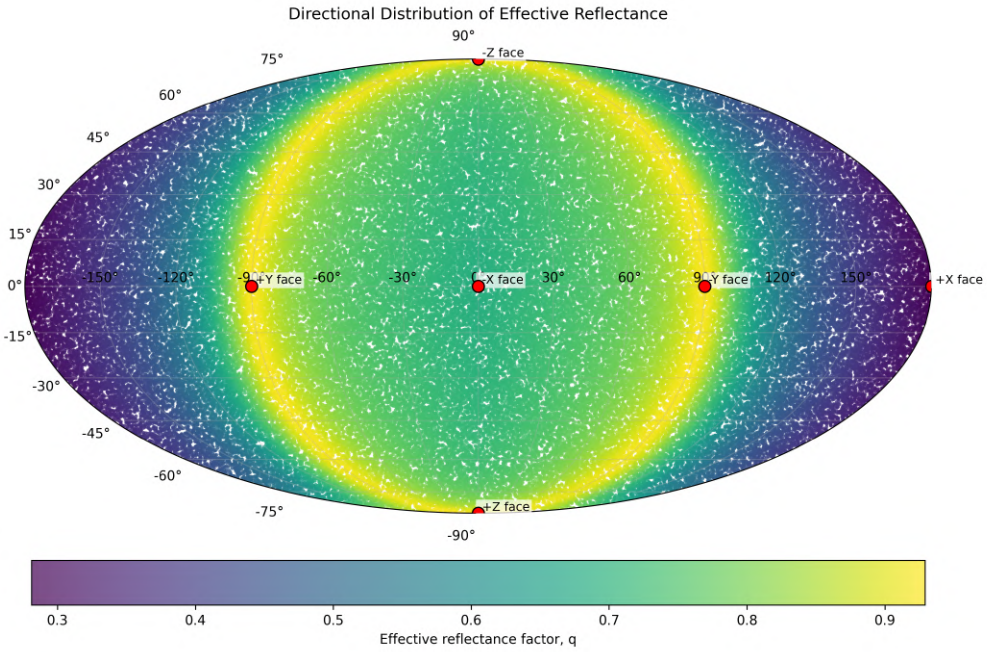


Figure 5: Mollweide projection of the effective spacecraft reflectance as a function of solar propagation direction. The principal illumination directions are marked to indicate the corresponding spacecraft faces.

Figure 5 provides a global view of the directional dependence of the effective reflectance. Each point represents one solar propagation direction, while the color scale indicates the corresponding value of q_{eff}. The marked principal directions provide a direct link between the Mollweide coordinates and the spacecraft faces. Directions located between two or three principal markers correspond to oblique illumination, for which multiple spacecraft faces contribute simultaneously to the effective reflectance.

8 Application to the Solar Radiation Pressure Model

The effective reflectance factor obtained from the optical analysis can be introduced directly into the simplified solar-radiation-pressure model used in this work,

$$F = \frac{F_s}{c} A_s (1 + q) \cos I.$$

Using the nominal Monte Carlo value, $q_{\text{nominal}} \approx 0.67$, the radiation pressure coefficient becomes $C_R = 1 + q = 1.67$. For a more conservative directional estimate, the upper tail of the Monte Carlo distribution may also be considered. The 95th percentile is $P_{95} = 0.904695$ which corresponds to $C_R = 1 + P_{95} \approx 1.905$

The solar-radiation-pressure loading depends not only on the effective reflectance factor, but also on the projected illuminated area. Figure 6 shows the relationship between the projected area and the effective reflectance obtained for the Monte Carlo illumination directions. The color scale represents the quantity $A_{\text{proj}}(1 + q)$, which is directly proportional to the radiation-pressure force in the simplified SRP model.

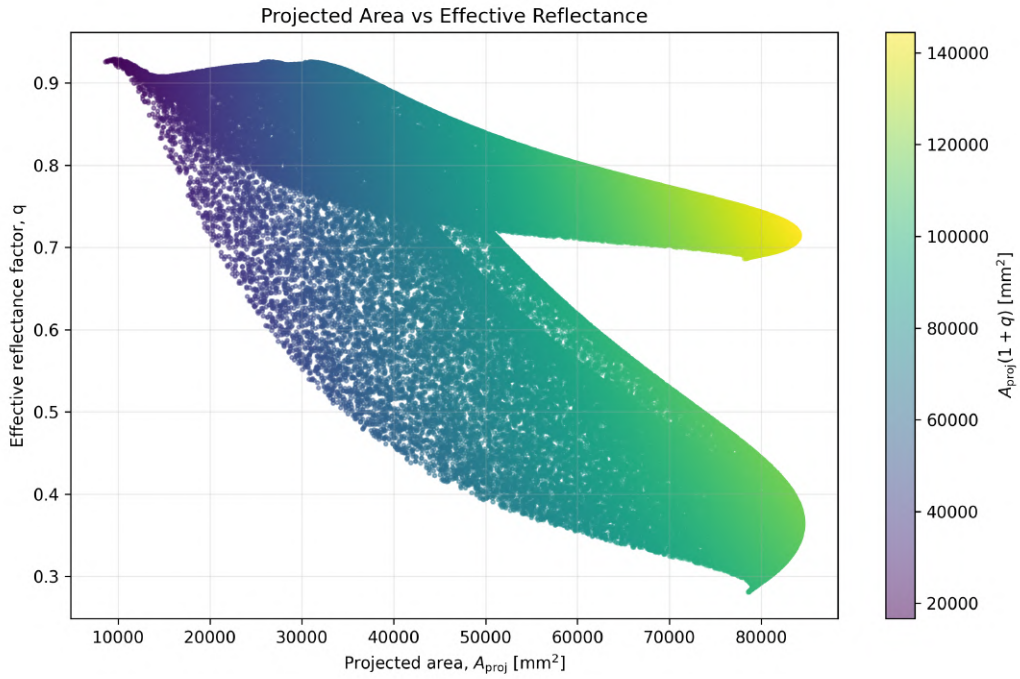


Figure 6: Projected illuminated area versus effective reflectance. The color scale represents the SRP loading factor $A_{\text{proj}}(1 + q)$.

The results show that the direction producing the highest reflectance does not necessarily correspond to the direction producing the highest SRP loading, since the projected illuminated area varies simultaneously with the illumination direction.

9 Limitations

The present analysis is based on a simplified optical representation of the spacecraft and therefore several limitations should be considered when interpreting the resulting reflectance factor.

First, the aluminium optical properties are represented using a common bulk aluminium optical dataset for all aluminium alloys included in the spacecraft model. In reality, the effective optical response may depend on alloy composition, surface finish, oxide layers, coatings, contamination and manufacturing processes.

The CMG-100 coverglass is represented using a constant refractive index and zero extinction coefficient over the considered wavelength range. This approximation does not include possible wavelength-dependent absorption, anti-reflective coatings, adhesive layers or the underlying solar-cell stack.

The spacecraft geometry is represented by a finite set of planar external surfaces with prescribed areas and outward normals. Local geometrical details, surface curvature and self-shadowing between spacecraft components are not included.

The Monte Carlo illumination model assumes uniformly distributed solar propagation directions over the unit sphere. This distribution is used to characterize the directional optical response of the spacecraft and does not represent a specific mission attitude profile.

Finally, the simplified SRP model uses a single effective reflectance factor and does not explicitly distinguish between specular and diffuse reflection components.

10 Conclusions

A numerical optical model was developed to determine the effective solar reflectance factor of the spacecraft as a function of illumination direction. The model combines Fresnel reflection, wavelength-dependent material properties, AM0 solar spectral weighting and projected-area weighting over the illuminated external spacecraft surfaces.

A Monte Carlo analysis using $M = 100000$ uniformly distributed solar propagation directions resulted in a mean effective reflectance of

$$\bar{q} = 0.667812,$$

with a standard deviation of

$$\sigma_q = 0.175791.$$

The corresponding median and upper 95th percentile were

$$P_{50} = 0.723773, \qquad P_{95} = 0.904695.$$

For use in the simplified solar-radiation-pressure model, a nominal reflectance factor of

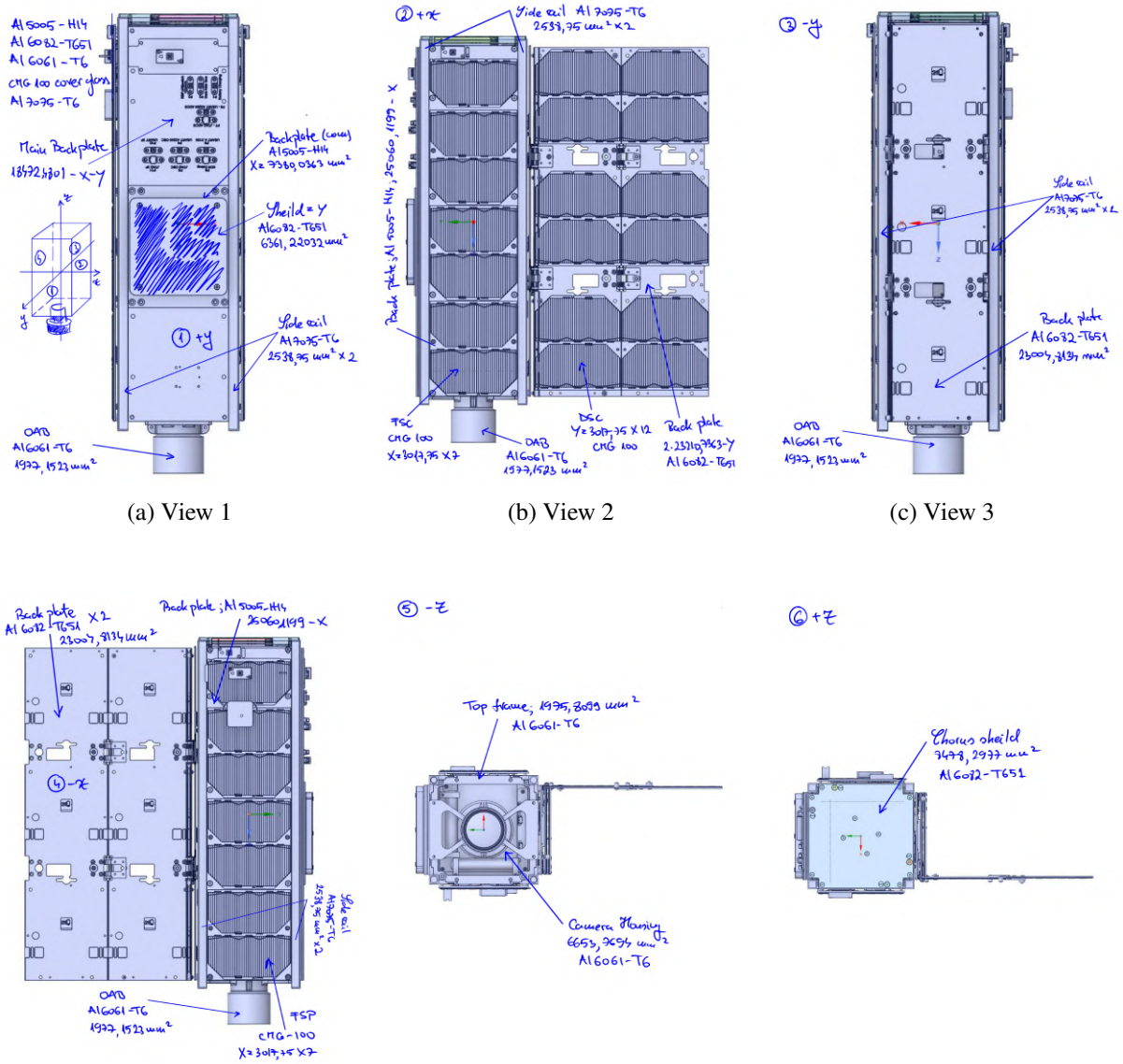
$$\boxed{q_{\text{nominal}} \approx 0.67}$$

is therefore proposed. In the formulation used in this work, this corresponds to an effective radiation-pressure multiplier of

$$C_R \approx 1.67.$$

A Spacecraft Surface Identification

The annotated spacecraft views used to derive the surface areas, material assignments and outward normal directions included in the optical model are provided in this appendix.



(a) View 1 (b) View 2 (c) View 3

(a) View 4 (b) View 5 (c) View 6

Figure 8: Annotated spacecraft views used for surface identification.